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Hybridisation model boundaries — Issue #37 candidate:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In formamide, $\text{H}-\text{C}(=\text{O})-\text{NH}_2$, a local Lewis count gives nitrogen three sigma bonds and a lone pair. You are also given that the

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approximately planar amide arrangement permits nitrogen-lone-pair donation into the adjacent carbonyl pi system, which requires a suitably aligned p-like orbital. Choose a model that accounts for both the sigma framework and delocalisation. Resolve the apparent conflict with an automatic 'four domains therefore sp^3 ' label and state the limits of your chosen hybridisation description.

Need the concept again?

When a hybridisation label stops being the answer

Conditions

- Molecule is formamide, $H-C(=O)-NH_2$.
- Local Lewis count at nitrogen is three sigma bonds and one lone pair.
- The supplied approximately planar arrangement permits nitrogen-lone-pair donation into the adjacent carbonyl pi system.

- The supplied donation model requires a suitably aligned p-like orbital.

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Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the formamide situation supplied in Q1, construct one qualitative representation

Representation

connecting the Lewis lone pair on nitrogen, the nitrogen orbital used for donation and the carbonyl pi system. Then construct the corresponding approximately 90-degree-twisted arrangement about the C–N bond. Preserve atom connectivity and distinguish changes in orbital alignment from changes in formal electron count. Do not claim these sketches alone measure the donation strength.

Conditions

- Use the formamide situation supplied in Q1.
- Compare an approximately planar arrangement with an approximately 90-degree twist about the C–N bond.
- Atom connectivity must be preserved.
- Formal electron count must be distinguished from orbital alignment.

Fixed connectivity: H-C(=O)-NH_2 in b
End-on view along C-N; centres proje

Connectivity

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Need the concept again?

When a hybridisation label stops being the answer

- The qualitative sketches must not be treated as measurements of donation strength.

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Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Representation

The benchmark supplies a qualitative model result: with atom connectivity fixed, twisting a planar amide arrangement toward 90 degrees reduces the nitrogen-to-carbonyl orbital alignment and the delocalisation contribution. Explain that result through orbital overlap. Assess whether 'the hybridisation label changes, so donation disappears' is a sufficient explanation, and identify which parts of your account are model assumptions rather than measured facts.

Conditions

- Atom connectivity is fixed.
- The benchmark supplies that twisting toward 90 degrees reduces nitrogen-to-carbonyl orbital alignment and the delocalisation contribution.

Fixed connectivity: H-C(=O)-NH_2 in b
End-on view along C-N; centres proje

Connectivity

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Need the concept again?

When a hybridisation label stops being the answer

- The explanation must distinguish model assumptions from measured facts.

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Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Compare two nitrogen sites: A is the amide nitrogen in $\text{H}-\text{C}(=\text{O})-\text{NH}_2$; B is the amine

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nitrogen in $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$. In the simplified introductory model, the intervening saturated CH_2 group has no continuous aligned p-orbital pathway. Develop a decision procedure combining local sigma/lone-pair counting, adjacency, orbital alignment and the model's validity conditions. Apply it to A and B, then state a counterfactual structural change that would require you to reassess your procedure. Do not equate spatial proximity with conjugation.

Need the concept again?

When a hybridisation label stops being the answer

Conditions

- Site A is the amide nitrogen in $\text{H}-\text{C}(=\text{O})-\text{NH}_2$.
- Site B is the amine nitrogen in $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$.
- In the simplified introductory model, saturated CH_2 provides no continuous

aligned p-orbital pathway.

- Spatial proximity alone must not be equated with conjugation.

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Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

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A student claims: 'Hybridisation uniquely determines bond angles and any deviation proves the hybridisation label wrong.' The benchmark supplies approximate observations: CH₄ tetrahedral about 109.5 degrees; NH₃ pyramidal about 107 degrees; H₂O bent about 104.5 degrees; all three commonly use an introductory sp³ description at the central atom. It also supplies approximately planar amide nitrogen with lone-pair delocalisation. Construct and test a qualified replacement claim using these observations, local domain counting, lone-pair effects and the amide boundary case. State what further evidence would be needed before treating a qualitative orbital picture as a unique physical account.

Need the concept again?

When a hybridisation label stops being the answer

Conditions

- Use the supplied approximate observations: CH₄ about 109.5 degrees, NH₃ about 107 degrees, H₂O about 104.5 degrees.
- All three are commonly introduced using an sp³ description at the central atom.
- The benchmark also supplies approximately planar amide nitrogen with lone-pair delocalisation.
- The replacement claim must state what further evidence is needed for a unique physical account.

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Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the two nitrogen sites A and B in Q4. State the conditions under which the local four-domain counting rule is a useful introductory classification, and the evidence that would make its automatic use inadequate. Distinguish recalling a rule from deciding whether its conditions hold.

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Need the concept again?

When a hybridisation label stops being the answer

Conditions

- Use Sites A and B as defined in Q4.
- Distinguish remembering the local four-domain counting rule from deciding

whether its assumptions hold.

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Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A neutral methyl radical, $\text{CH}_3\cdot$, is supplied as approximately planar, with the unpaired electron in a p-like orbital perpendicular to

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the three C–H sigma directions. Translate that supplied model into an orbital sketch and assess whether 'three sigma bonds always leave an empty p orbital' follows. Keep the given model distinct from a universal claim.

Need the concept again?

When a hybridisation label stops being the answer

Conditions

- The benchmark supplies neutral $\text{CH}_3\cdot$ as approximately planar.
- The unpaired electron is supplied as occupying a p-like orbital perpendicular to the three C–H sigma directions.
- The supplied model must be kept distinct from a universal claim.

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Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Two qualitative descriptions of formamide are proposed: a completely localised pyramidal nitrogen lone pair, and an approximately planar nitrogen arrangement with possible lone-pair donation into the carbonyl pi system. Propose observations that could discriminate between them, explain the warranted inferences, and identify at least

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When a hybridisation label stops being the answer

one inference that geometry alone cannot establish.

Conditions

- Compare a completely localised pyramidal nitrogen-lone-pair description with an approximately planar nitrogen arrangement allowing possible donation into the carbonyl pi system.
- At least one proposed inference must be identified as not establishable by geometry alone.

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Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies a simplified torsional model in which the relevant p-orbital overlap factor varies as $\cos(\theta)$, where θ is the angle between their alignment directions. Compare $\theta = 0, 60$ and 90 degrees, then assess the claim that the molecule's total energy must therefore vary as $\cos(\theta)$. Separate execution of the given model from a justified claim about the physical system.

Representation

Fixed connectivity: H-C(=O)-NH_2 in ball-and-stick model
End-on view along C-N; centres projected

Connectivity

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Need the concept again?

When a hybridisation label stops being the answer

Conditions

- The supplied simplified torsional model defines the relevant p-orbital overlap factor as $\cos(\theta)$.
- θ is the angle between the relevant alignment directions.
- Compare $\theta = 0, 60$ and 90 degrees.
- Execution of the supplied overlap model must be separated from claims about total molecular energy.

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Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: When a hybridisation label stops being the answer

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Local counting, orbital alignment and model-scope decisions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A proposed teaching rule says, 'Count sigma bonds and lone pairs, assign a hybridisation label, and the label explains every shape and reactivity difference.'

Audit this rule against NH_3 , allene, formamide and the separated amine site $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$. Produce a qualified decision framework, identify where local counting is useful, and state what evidence or model is needed beyond it.

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Need the concept again?

When a hybridisation label stops being the answer

Conditions

- Audit the proposed rule against NH_3 , allene, formamide and $\text{H}-\text{C}(=\text{O})-\text{CH}_2-$

NH₂.

- The final framework must identify where local counting is useful and where additional evidence or a model is needed.

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